Open problems:
Inconsistent convergence in Hilbert space

Open problems raised by the technical brief in this directory, and by its Lean formalization (see `formalization-log.tex`).

From the brief

1. **Unbounded observables in general.** The introduction states that the domain of an unbounded self-adjoint operator cannot be the whole space, and that every unbounded operator is "not continuous in its own domain", e.g. for the average position. Only the single example $\langle N \rangle$ on ℓ^2 is formalized; the general statements (Hellinger–Toeplitz, discontinuity of $\psi \mapsto \langle \psi | A | \psi \rangle$ for every unbounded A) are not. (Brief, Sec. 1.)

2. **Schwartz topology prevents it.** "A topology that keeps track of all moments, like the Schwartz topology, would not allow this kind of convergence." That $\langle N \rangle$, and expectations of polynomial observables in general, are continuous on $S(\mathbb{N})$ is not proved. (Brief, Sec. 3, last paragraph.)

3. **A criterion of physical convergence.** The brief concludes that the ℓ^2 topology "is too weak to take into account all conditions for a meaningful criterion of physical convergence". Which topology is the right one, and whether it is the Schwartz topology, is left open. (Brief, Sec. 4.)

From the formalization

1. **The space $S(\mathbb{N})$.** The Schwartz sequence space and its topology are not constructed. Only unboundedness of the seminorm $\sum_n n^2 |e_n^j|^2$ is proved, not that e^j (or b^j) has no limit in $S(\mathbb{N})$. (Log, "Differences from the brief", "No limit in $S(\mathbb{N})$".)

2. **The number operator.** $\langle N \rangle$ is a weighted sum $\sum_n n\,|x_n|^2$ on all of ℓ^2, with a default value outside the domain of N. The unbounded self-adjoint operator $N = a^\dagger a$, its domain, and the agreement of $\langle x | N | x \rangle$ with the weighted sum on that domain are not formalized. (Log, "Differences from the brief", third item.)

3. **The oscillator.** $\ell^2(\mathbb{N})$ with $|n\rangle = \delta_n$ is only read as the harmonic oscillator. The unitary identification with $L^2(\mathbb{R})$ through the Hermite functions, under which $S(\mathbb{N})$ becomes the Schwartz space of functions, is not formalized. (Log, "Setting".)